

Education

Università degli Studi di Catania

M.S. IN COMPUTER SCIENCE: ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Current

- Relevant Coursework: Data Analysis, Cloud Computing, Natural Language Processing (NLP), Social Media Data Analysis, Deep Learning.

Università degli Studi di Catania

B.S. IN COMPUTER SCIENCE: ARTIFICIAL INTELLIGENCE & ROBOTICS - GRADE: 106/110 (≈ 3.9 GPA)

Sep. 2025

- Graduated on time (less than 40% of peers succeeded) with a top score significantly above the cohort average.
- Best Marks: Database, Operating Systems, Computer Networking, Object-Oriented Programming (C++) (4.0 GPA overall).

Università degli Studi di Catania

B.S. IN POLITICAL SCIENCE GRADE: 108/110 (≈ 4.0 GPA)

Feb. 2023

- Gained strong analytical foundation in statistical analysis and inferential statistics applied to social data.

Thesis Projects

B.S. in Computer Science

UNDERSTANDING EGOCENTRIC VIDEOS IN THE SURGERY DOMAIN

Sep. 2025

- Applied computer vision and machine learning techniques to examine a dataset of over 17 hours of egocentric surgical videos.
- Identified novel insights from more than 100 annotated frames of hands and instruments, not previously reported in related studies, obtaining recognition performance of 50–60% and highlighting the complexity of open surgery video analysis and the need for domain-specific adaptation.

B.S. in Political Science

ARTIFICIAL NEURAL NETWORKS APPLIED TO THE STUDY OF "BRAIN DRAIN"

Feb. 2023

- Applied an ANN to over 58,400 AlmaLaurea survey responses to analyze socio-demographic and career characteristics of Italian graduates, revealing that 10.5% migrated internally and 5.7% abroad.
- Attained 85.4% predictive accuracy in forecasting migration tendencies based on 13 key input variables using AI methodologies.

Side Projects

Natural Language Processing | Kaggle Notebook

SEMANTIC ALIGNMENT OF EMOJIS IN TRANSFORMER MODELS

2026

- Developed a semantic alignment pipeline using the CardiffNLP Twitter 100M corpus, implementing emoji deduplication and masking.
- Fine-tuned a *twitter-roberta-base* model using a Triplet strategy and Multiple Negatives Ranking Loss (MNRL) to resolve "Context Domination".
- Achieved a 54.8% Top-1 and 90.7% Top-5 Accuracy in an Information Retrieval task, demonstrating advanced semantic clustering.

Social Media Data Analysis | GitHub

ASPECT-BASED SENTIMENT ANALYSIS & LLM AUTO-LABELING

2026

- Designed an end-to-end pipeline to extract business intelligence from Italian banking reviews using Playwright for scraping.
- Implemented a hybrid labeling methodology using Gemini 2.0 Flash to generate a BIO-formatted ground truth, bypassing manual annotation costs.
- Fine-tuned an LCF-ATEPC architecture on an Italian BERT backbone, achieving 87.6% accuracy in sentiment classification.

Deep Learning | GitHub

IMAGE RESTORATION: UPSCALING AND DENOISING

2026

- Adapted a Generative Adversarial Network (SRGAN) to enhance image quality by combining 4x upscaling with noise removal.
- Improved visual clarity and texture recovery over standard models by fine-tuning the network on noisy images.

IoT & Machine Learning | GitHub | Slides

BOARDGUARD: EDGE AI & IOT SECURITY

2025

- Developed and deployed an Edge AI model on ESP32 to identify boat theft (unauthorized towing) by analyzing real-time data from motion and speed sensors.
- Designed a secure two-layer encryption protocol for a LoRaMesh network, ensuring private data transmission and communication across a decentralized fleet of vessels.

Database Systems

DATABASE DESIGN & OPTIMIZATION

2025

- Architected and optimized a MySQL relational database, executing entity partitioning and resolving redundancies for high-performance queries.
- Populated records using Python's Faker library and enforced complex business rules via custom SQL Triggers.

Hackathon by NeoData | GitHub

IDEATOR AND DEVELOPER OF FLYNOW — SECOND PLACE WINNER

2024

- Led a 3-person team to build an AI-powered travel planning prototype using OpenAI during a 48-hour competitive hackathon.
- Streamlined flight and accommodation search workflows, theoretically reducing user planning time by 60–70%.

Extracurricular Activity

Quality Development | GitHub

UNIVERSITÀ DEGLI STUDI DI CATANIA

Nov. 2025

- Course focused on software engineering best practices, covering UNIX shell scripting, advanced Git workflows, and Object-Oriented Design (SOLID principles).
- Gained hands-on experience in software testing and automation, implementing unit tests with *pytest* and setting up CI/CD pipelines for code linting and continuous deployment.
- Applied theoretical concepts by developing a Pacman clone (Pyc-Man) in Python, managing the repository via conventional commits, pull requests, and automated GitHub Actions.

Intelligent Agents and Machine Learning

ORGANIZED BY AITHO

July 2025

- Completed 75 hours of intensive coursework covering 7 architectures of intelligent agents and their applications in AI and machine learning.
- Explored and applied 3 practical tools/frameworks including LangChain, LangGraph, and Ollama to develop multi-agent local projects.

AI and Robotics: Fundamentals and Applications with Micro-controllers

ORGANIZED BY ST-MICROELECTRONICS & UNI-CT

June 2024

- Completed a 24-hour course on AI and robotics fundamentals with focus on micro-controller applications.
- Passed the theoretical exam with a perfect score of 100%, demonstrating full mastery of the core concepts.

Qiskit Fall Fest 2024

ORGANIZED BY IBM & UNICT

Nov. 2024

- Attended sessions introducing quantum programming concepts using Qiskit.
- Successfully completed foundational quantum programming assessments, earning the course certificate.

Skills

TECHNOLOGIES & TOOLS

- **Languages & Software Engineering:** Python, C/C++, SQL, Bash | Git/GitHub Actions, Docker, Linux, CI/CD, Pytest, SOLID
- **Artificial Intelligence (ML, NLP, CV):** PyTorch, TensorFlow (TinyML), Scikit-learn, Hugging Face, spaCy, LangChain, GenAI APIs
- **Data Engineering & Scraping:** Pandas, NumPy, Matplotlib, Playwright, Selenium, Regex
- **IoT, Edge & Cloud Services:** ESP32, LoRaMesh, AWS

Additional Experience

Professional Firm

JUNIOR ACCOUNTANT

2022-2023

- Managed financial data for 100+ accounts and developed an automated script to streamline invoice follow-ups, improving efficiency by 60%.